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Professional Summary

Fourth-year Informatics Engineering student focused on low-level programming, embedded systems and hardware security. Experience includes a TPM 2.0 research internship at IKERLAN and hands-on IoT, networking and embedded projects. Seeking a master's in IoT/embedded systems and a working-student role in Madrid.

Education

- **University of the Basque Country (UPV/EHU)** — Donostia, Spain
  B.Sc. in Informatics Engineering; Fourth-year student (Expected graduation: May 2027) — *Sep 2023 – May 2027*
  - **Academic record**: 8.64/10 average; 9 *Matrícula de Honor* distinctions (highest grade). Relevant coursework: Computer Architecture, Operating Systems, Computer Networks, Distributed Systems, Applied Cryptography, Systems Auditing & Forensics, Vulnerability Analysis, Parallel Computing, High-Performance Processors.
- **Erasmus+ Exchange — Université de Pau et des Pays de l'Adour (UPPA)** — Anglet, France
  M1 SIGLIS — Computer Science for Industrial Logistics and Services — *Dec 2026 – May 2027*
  - **Erasmus coursework**: DevOps Methodology 2, Enterprise Integration Patterns, Cybersecurity, Green IT, Continuous Improvement, Technical English.

Experience

- **IKERLAN** — Hernani / Arrasate, Spain
  Student Intern — Embedded Cybersecurity Systems — *Feb 2026 – Jun 2026*
  - **TPM 2.0 Analysis**: Analyzed TPM 2.0 operation (PCRs, measured boot, attestation) and implemented direct SPI communication bypassing OS drivers to optimize embedded access.
  - **Internal Framework (Chaser)**: Developed a TPM-specific module for the company's modular Chaser framework, adding automated capture, filtering and key-extraction workflows; contributed to the framework core and its improvement.
- **Polymat** — Donostia, Spain
  Research Intern — *Jun 2022 – Aug 2022*
  - **Laboratory Research**: Hands-on laboratory protocols, experimental documentation and reproducibility practices.
- **Cruz Roja Juventud (Spanish Red Cross Youth)** — Donostia, Spain
  Volunteer & Council Representative — *Sep 2022 – Present*
  - **Multi-Level Leadership**: Member of Donostia and Gipuzkoa councils with national assembly representation; coordinated volunteer activities and social impact projects.

Technical Skills

- **Programming Languages**: C, C++, Assembly (basic), Java, Python, Shell/Bash, SQL
- **Systems & Embedded**: Embedded Systems, Microcontrollers, Memory Management, Hardware Interfaces, Raspberry Pi, TPM 2.0, Secure Boot, Attestation, SPI
- **Infrastructure**: Linux/Unix, Proxmox, Docker, ZFS, Cloudflare Tunnel, Home Assistant, Keycloak, LDAP, Caddy
- **Development Tools**: Git, GCC/Clang, OpenMP, Pthreads, devkitPro, Maven, Saleae Logic, Enterprise Architect
- **Networks & Protocols**: MQTT, TCP/IP, DNS, VPN, Zigbee, Z-Wave
- **Specialized Areas**: Computer Architecture, Embedded Security, IoT Development, System Optimization, Agile Development

Technical Projects

- **Smart Home & Homelab Infrastructure** Self-hosted
  Home Assistant · MQTT · Zigbee · Z-Wave · Proxmox · Docker · ZFS · Keycloak · Cloudflare Tunnel
  - **Smart home**: Deployed Home Assistant on a Raspberry Pi, integrating WiFi, Zigbee and Z-Wave devices through MQTT and building automations for monitoring and control.
  - **Homelab**: Run a Proxmox-based homelab with Docker and ZFS, hosting self-hosted media, mail and cloud services.
  - **Identity**: Centralized authentication with Keycloak backed by LDAP and PostgreSQL, providing a single sign-on layer for self-hosted services.
  - **Remote access & hardening**: Expose services through Cloudflare Tunnel and VPN, terminate TLS at Caddy, and isolate workloads across containers and VMs.
- **Nintendo DS Game Development** Bare-metal C
  C · devkitPro · Hardware interfaces
  - **Platform**: Built a complete Nintendo DS game in C with the devkitPro toolchain, targeting real hardware constraints: limited RAM, dual screens and custom input.
  - **Graphics**: Implemented direct hardware manipulation, custom graphics rendering and dual-screen coordination without a game engine.
  - **Optimization**: Kept memory usage and rendering within the console's limits, prioritising responsive controls and stable behaviour.

Honors & Awards

- **IK3RL4N CTF — 1st Place (Team 0xHunter$)** Bilbao, Spain
  University-level cybersecurity competition; Organized by IKERLAN *Nov 2025*
  - **Technical skills**: Reverse engineering, web exploitation, cryptography, and digital forensics.
- **SWERC — Contestant** Lisbon, Portugal
  ICPC Southwestern Europe Regional Contest; team-based algorithmic problem solving *2024*
- **Becas Europa XVIII — Top 200 pre-university students in Spain** Spain
  Leadership and academic program *2022 – 2023*

Languages

English: C2 (Proficiency) **French**: C1 (DALF) **Spanish**: Native **Basque**: Native

Additional

Driving licence B